25 Years of SMOTE: What Has Been Done and How Did It Benefit Society?

Data Section, Stand-Alone Draft

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This document typesets only the Data section of the survey with the JAIR template, so that the co-authors can read it in the journal's layout before it is merged into the full manuscript. Section numbering starts at 1 here and will change on merging; the reference list holds only the works this section cites.

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1 How the SMOTE Literature Uses Data

A reader who wants to compare two SMOTE variants, or to choose a benchmark for a new one, needs answers to questions that the papers rarely answer together. Can the numbers that two papers report on a dataset of the same name be compared? How broad is the evidence behind the benchmarks the field relies on? How difficult, beyond the imbalance ratio, are the datasets that can be checked? When a paper reports that its method beats SMOTE, on what kind of data does that happen, and can a reader check it? This section answers these questions from the 1,654 dataset entries and the comparisons with SMOTE that we extracted from the 76 papers of our corpus: 835 under one fixed metric per paper (834 per dataset and one average over datasets) and 2,497 over every metric the papers report. Three of them (ENN, AdaBoost and one review) contain no evaluation on data, so counts below use the 73 papers with an evaluation on data; Tomek links is kept because it illustrates its method on one synthetic two-class set.

The central observation is that the evidence for a new oversampler is only as interpretable as the description of its data, and in this literature that description is often incomplete: the same name can denote different tasks, many modifications are not described well enough to be repeated, and 14 of the 57 papers that use SMOTE as a baseline give no per-dataset pair of scores against it. Where per-dataset results exist, the proposed method beats plain SMOTE in about four of five decided comparisons. These wins are not more frequent on data the authors modified, but they are more frequent on data whose minority examples are borderline or isolated, and they are much less consistent on precision and specificity than on the other metrics.

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1.1 Extraction Procedure

For every paper we read the full text and any supplementary file in our collection, and recorded each dataset as printed, including spelling, with its group in the paper, source repository, size, number of features and classes, minority and majority counts, and the imbalance ratio where the paper prints them, together with the table and page they come from. For each dataset we also recorded how it reached the experiment: *original* (used as distributed, with its native classes), *repository-defined* (a two-class split published by a repository such as KEEL or imbalanced-learn, used as is), *modified* (the authors chose the minority classes, merged or dropped classes, set a threshold, undersampled or injected noise), or *synthetic* (generated by the authors). Where a paper did not state its source, two fixed rules were applied and flagged: names that follow the KEEL file convention (for example `ecoli-0-1_vs_5`, `yeast4`) were labelled repository-defined, and common names in multiclass papers with no stated change were labelled original. Two further rules were fixed before counting: a set that is two-class in its source and is used without comment was labelled original, and a set from which the authors removed rows (for example, records with missing values) or attributes was labelled modified.

For the results we copied, per dataset, the printed score of the proposed method and of SMOTE from one results table selected by a fixed rule: G-mean, else AUC, else F-measure, else accuracy; the classifier highlighted in the abstract, else the first reported; the unperturbed condition. We recorded values rather than wins so that any definition of a win can be applied later. From the same table setting we then copied every other metric that the paper reports for both methods (2,497 dataset-by-metric values in 44 papers); the fixed-metric values were copied a second time in this pass, independently, and agree on all 839 rows that both copies contain. Plain SMOTE was distinguished from SMOTE embedded in a multiclass or ensemble wrapper (Static-SMOTE, OVO-SMOTE (Fernández, López, et al. 2013), SMOTE-all, SMOTE+AdaBoost). Values printed in a paper are never overwritten; corrections against the source repository are stored beside them with their justification. Every extracted value was then checked against the PDF a second time: a value was accepted when the dataset name and its numbers appear on the same line of the cited page, and every table that is printed as an image or whose text layer is garbled was read from the rendered page instead. All 1,654 dataset entries and 835 fixed-metric comparisons passed this check; the values of the other metrics were read once, from the text layer or the rendered page.

1.2 The Benchmark Landscape

The 73 papers with an evaluation on data list 1,654 dataset entries. One of them, the single toy example in Tomek links, is a description rather than a name and is set aside here, leaving 1,653 named entries. These reduce to 771 distinct names once names are lower-cased, text in parentheses is dropped, and underscores and spaces are treated alike. Treating spellings that differ only in punctuation as one dataset (`page-blocks`, `pageblocks` and the singular `page-block`) leaves 680 datasets, and grouping each numbered or split version with its main set (for example `yeast4` or `glass-0-1-6_vs_2` with yeast and glass), while keeping non-standard names such as `glass identification` apart, leaves 459 source families. Most datasets appear once: 463 of the 680 occur in a single paper. Counted as datasets, ecoli appears in 29 papers, yeast in 22 and glass in 19; counted with their splits, the ecoli family appears in 42 papers, the yeast family in 39 and the glass family in 35, about half of the 73. Half of all uses of a family fall on 37 of the 459 families. Names whose printed number of features contradicts their main set are kept in its family when they are the same source with a different composition (for example `abalone8`, `oil1`) and kept apart when they are a different data set that shares the prefix (for example `car1`, a microarray set with 9,182 features).

Splits cut from one source are not copies of each other. Among the verifiable two-class KEEL files (Section 1.5), the 14 yeast splits range in imbalance ratio from 2.5 to 41.4 and in noise index from 0.29 to 0.86, and the 19 ecoli and 13 glass splits from 0.12 to 0.69 and from 0.24 to 0.77. They do share the source's attributes (7 or 8 for yeast, 6

or 7 for ecoli, 9 for glass), its domain and often its examples: `yeast4` and `yeast-0-5-6-7-9_vs_4` use the same 51 minority examples and differ only in which classes form the majority.

Outcomes on splits of one source are also related. In the eleven papers that compare with plain SMOTE on several splits of the same source (282 per-dataset comparisons, 262 same-family pairs; families by the rule in the evidence note), whether the proposed method wins does not cluster by family (pairs agree on win, tie or loss in 0.60 of cases, as under random relabelling), but the size of its margin does: the within-paper ranks of the margin differ by 0.29 on average for two splits of one source against 0.34 when family labels are permuted within each paper ($p = 0.005$, 10,000 permutations; leaving out one paper at a time gives p between 0.002 and 0.048).

Splits cut from one source vary the task (its imbalance ratio and where the minority examples lie) but not the source (its domain, attributes and dimensionality), and within the papers that evaluate several splits of one source, the margin over SMOTE tends to be slightly more alike within a family than between families. Papers should therefore report the number of source families alongside the number of datasets, and should not treat each split as independent evidence.

What the reader should do. Report the source family of each dataset alongside the count, so that readers can see how many independent sources stand behind a claim, and keep splits of one source together when testing for significance.

Candidate benchmarks for a cross-family comparison. Table 1 lists 18 datasets considered as benchmarks for comparing method families: the 11 datasets we set out to examine and the datasets that meet one of two written rules. The first rule takes every dataset used by at least ten papers (33 datasets). The second takes datasets used by three to nine papers whose every printed size is at least 10,000 instances, a size that no entry of the first group reaches (7 datasets). The groups follow from these two rules on use and printed size only and say nothing about overlap, noise, sparsity, feature type or any other property. IR is not used, because its definition differs between papers. No dataset used by three or more papers is printed with 100 or more features every time it is reported; this is a property of the rule, not evidence that the corpus contains no high-dimensional data sets.

Family counts follow the same rule for every dataset. The dataset `shuttle` is printed with four different sizes (2,175, 14,500, 43,500 and 57,999 instances) and once without a size. Two pairs of papers print matching sizes and class counts; the other versions are single reports, and OVO-SMOTE prints five classes in one table and seven class counts in another. The versions are therefore kept apart and not pooled.

The frequently used benchmarks are small: among the entries that print a size, half have 800 or fewer instances and half have nine or fewer features. Names printed with 10,000 or more instances in every paper are used by three to nine papers each, so a comparison across method families that wants to cover data size has to look beyond the frequent set.

Limits of the evidence. Size, dimensionality and the number of classes are printed for 80.2%, 75.9% and 77.4% of the 1,653 named entries, and class proportions in some form for 56.8%. A measured class overlap exists for 36 entries in two papers, on two incompatible scales, and the curated evidence contains no statement about sparsity. What the papers print therefore cannot show which ranges of overlap, noise or sparsity the benchmarks cover; the candidates in Table 1 are chosen on use and size only. Section 1.5 measures these properties directly for the entries whose files can be identified.

1.3 How Datasets Are Chosen and Modified

Table 2 shows that modification by the authors is common: 331 of 1,654 dataset entries were changed by the authors themselves, and for 199 the paper does not let a reader tell how the data were obtained. Eleven papers give no reason for their choice of datasets or only call them widely used.

Table 1. Data Sets Considered as Benchmarks for Cross-Family Comparison. F: used by ten or more papers; D: used by three to nine papers, every printed size at least 10,000 instances; V: needs verification. Papers count the dataset (spellings that differ only in punctuation are one dataset); family papers include the numbered and split versions of the dataset. Values are as printed and differ between papers. [a]Four printed versions, not pooled.

Dataset	Group	Papers	Family papers	Instances	Size printed	Classes	Modified
ecoli	F	29	42	301–336	25/31	2, 4, 5, 8	8/31
page-blocks	F	22	34	545–5,476	17/22	2, 4, 5	6/22
yeast	F	22	39	483–1,485	16/22	2, 6, 8, 9, 10	4/22
pima	F	21	21	590–769	19/23	2	2/23
vehicle	F	21	30	713–940	19/23	2, 3, 4	10/23
haberman	F	20	20	252–306	19/22	2	2/22
glass	F	19	35	167–214	18/21	2, 6	7/21
balance	F	16	16	625	14/16	3	3/16
wine	F	16	16	130–178	15/18	2, 3	6/18
satimage	F	15	16	6,430–6,435	11/15	2, 6, 7	8/15
letter	D	9	9	11,057–20,000	9/9	2, 26	6/9
nursery	D	7	7	12,690–12,960	7/7	2, 4, 5	3/7
mammography	D	5	5	11,183	4/5	2	0/5
adult	D	3	3	32,562–48,842	2/3	2	1/3
avila	D	3	4	10,430–20,850	3/3	2, 11	1/3
default	D	3	3	30,000	2/3	2	0/3
shoppers	D	3	3	12,330–12,331	3/3	2	0/3
shuttle	V	7	15	2,175–57,999[a]	6/7	2, 5	1/7

Table 2. How the Papers Choose and Obtain Their Data. Top: paper level, 73 papers with an evaluation on data. Bottom: dataset entries.

Item	Count
Reason given for the choice of datasets (papers)	
Own reason (diversity, difficulty, application, …)	48
Cites prior work	14
“Widely used” only	3
Not stated	8
Benchmark list adopted from earlier work, declared (papers)	20 (wholesale 11)
Binarised, minority rule not stated (papers)	8 of 37
How each dataset entry reached the experiment	
Original	534
Repository-defined (KEEL, imbalanced-learn)	470
Modified by the authors	331
Synthetic	120
Cannot be told from the paper	199

Modification is not a fault in itself: binarising a multiclass set or raising the imbalance ratio are legitimate ways to probe a method. What matters for the reader is whether the modification is declared where a reader

Table 3. Imbalance-Ratio Definitions Printed in the Papers

Definition	Formula	Used by (examples)
Largest over smallest class	$N_{\max}/N_{\min}$	MDO (Abdi and Hashemi 2016), CBHSID (Palli et al. 2022), MKC-SMOTE (Wang and Awang 2024) (its *IR*)
Majority over minority (binary wording)	$N_{\mathrm{maj}}/N_{\min}$	OVO-SMOTE (defined for the binary case, applied to multiclass sets)
Percentage minority	$100\,N_{\min}/N$	RUSBoost (Seiffert et al. 2010)
Per-class imbalance degree	$\sum_{q\neq i} N_q/((C-1)N_i)$	OREM-M (Zhu, Liu, et al. 2023) (overall IR only cited, not printed)
Summed pairwise multiclass IR	$\sum_{i<j}(N_i/N_j - 1)$	SMOM (Zhu, Y. Lin, et al. 2017), GDHS (Yan et al. 2025), MKC-SMOTE (IR_{overall})
Imbalance degree *Id*	distance from uniform (Ortigosa-Hernández et al. 2017)	SHSampler (Zheng et al. 2022) (cited, no formula printed)
Minimum class prior	$N_{\min}/N$	DSRBF (Fernández-Navarro et al. 2011)
Coefficient of variation	$\mathrm{sd}(N_i)/\mathrm{mean}(N_i)$	COG-OS (Wu et al. 2007)
Proportion pair	e.g. 0.04:0.96	MWMOTE (wording: minority to majority)
Not stated (values equal $N_{\max}/N_{\min}$)	n/a	MC-CCR (Koziarski, Woźniak, et al. 2020), MC-RBO (Krawczyk et al. 2020)

looks for it. A KEEL file name such as `yeast-0-5-6-7-9_vs_4` declares its split; a paper that prints only "Yeast" after choosing its own minority classes does not.

Modifications differ in how completely they are described. Of the 331 modified entries (39 papers), 177 in 30 papers change the class structure, most often by keeping one class against the rest (84 entries), and 147 in 9 papers raise the imbalance ratio on purpose, almost always by removing minority examples at random (131 entries). A purpose for the change is stated in 15 of the 39 papers. For 93 of the 331 entries, information needed to repeat the change is missing: the characteristics of the modified set are not printed for 55, the target imbalance ratio for 18 and the minority class for 15. At the paper level, 8 of the 37 papers that binarise data do not state which class becomes the minority. Entries modified without printed characteristics cannot be matched to any public file, and 28 of them are therefore left out of the difficulty measurement (Section 1.5).

1.4 The Same Name Is Not the Same Data

Of the 192 datasets that at least two papers print with a size (spellings that differ only in punctuation counted as one dataset), 90 appear with more than one size: thyroid with 7,200 instances in five papers and 720 in four, dermatology with 358 (the KEEL version) in eight and 366 (UCI) in two, ecoli with 336, 327, 310 and 301, and pima with 768, 769, 635 and 590. Some of these differences are declared subsamples; the name alone does not tell which. MWMOTE (Barua et al. 2014) illustrates the problem inside a single paper: its Satimage (6,435 instances) and Statlandsat (4,435) are the full Statlog Landsat set and its training file, counted as two of its twenty datasets, and its common names are ambiguous in both directions (its Ecoli matches the KEEL file `ecoli1` exactly, while its Yeast matches no published split).

Imbalance ratios add a second layer. Table 3 lists the definitions found in the corpus; some are not convertible into one another, and some papers print IR without a formula.

Errors also travel with shared benchmark tables: MC-CCR and MC-RBO both print an IR of 1.17 for vehicle, a value no definition produces from its class counts.

What the reader should do. Compare results across papers only after mapping each name to its exact file and class split, and report the IR formula together with its values.

1.5 How Difficult Are the Datasets That Can Be Checked?

The papers rarely describe how hard their data are beyond the imbalance ratio. A numerical overlap measure is printed for 36 of the 1,653 named entries, all in two papers, and noise or borderline examples are described for 5; three of 72 papers report such a property of their original data (Section 1.2). The literature on imbalanced learning, however, attributes much of the difficulty of these problems to characteristics of the data other than the imbalance ratio: overlap between the classes, noisy and borderline examples, small disjuncts and a shortage of minority examples (Jo and Japkowicz 2004; López et al. 2013; Napierala and Stefanowski 2016). We therefore measured these characteristics ourselves on the entries whose data can be identified. An entry counts as verifiable when its name matches a public file and the printed number of instances (and of minority examples, when printed) equals that file's, or when the paper prints no size, the name matches a public file and the authors did not modify the data. The files come from KEEL (Alcalá-Fdez et al. 2011), the UCI repository (Kelly et al. 2026) (through the UCI++ collection (Garcia n.d.), the UCI files distributed with Weka and single-file copies), PMLB (Romano et al. 2022), imbalanced-learn (Lemaître et al. 2017) and the PROMISE defect data (Jureczko and Madeyski 2010). A name is looked up under the other published names of the same data set (liver disorders is `bupa`, contraceptive method choice is `cmc`), but only the printed size decides whether a file is accepted. This holds for 864 entries from 65 papers, which use 172 files. Entries modified by their authors without printed characteristics (35) are left out, because the data actually used cannot be reconstructed.

Each two-class file, each binarisation that a paper describes precisely enough to reproduce, and each imbalanced-learn task defines one task: 151 tasks, used by 594 entries in 45 papers. One-vs-rest tasks that we built from multiclass files (305) are not tasks any paper evaluated and are reported apart. Every minority example is labelled safe, borderline, rare or outlier from the classes of its five nearest neighbours (Napierala and Stefanowski 2016), with distances measured by HVDM (Wilson and Martinez 1997). The noise index of a task is the share of minority examples that are borderline or outliers; we report it as a continuous value because we found no published threshold for it. Three flags use published thresholds: small disjuncts when at least 20% of minority examples are rare and outliers when more than 20% are outliers (Napierala and Stefanowski 2016), and small sample when there are fewer than ten minority examples per feature (Jain et al. 2000). The imbalance ratio $N_{\text{maj}}/N_{\text{min}}$, computed from the files, is grouped into the bands 1.5 to 3, 3 to 9 and above 9 (Fernández, García, et al. 2008) and is not used as a difficulty flag.

Table 4 shows that the most common property of the binary tasks is a small sample: 103 of 151 have fewer than ten minority examples per feature. The flags combine in six profiles: small sample alone (67 tasks), small sample with outliers (21), all three flags (14), outliers alone (3), small disjuncts with a small sample (1) and none (45). The noise index separates tasks that the flags do not (Figure 1a): among the 45 tasks without a flag it ranges from 0 to 0.84, with 13 tasks at 0.5 or above (haberman 0.84, the numeric German credit data 0.81, contraceptive 0.79 and bupa 0.79 among them) and 11 below 0.1. Its median is 0.37, 0.43, 0.33 and 0.55 in the imbalance-ratio bands up to 1.5 (4 tasks), 1.5 to 3 (31), 3 to 9 (29) and above 9 (87), and its rank correlation with the imbalance ratio is 0.18.

The measurable part of the corpus is also low-dimensional (Figure 1b). The printed number of features has a median of 9 and half of the entries lie between 4 and 10 features. The 59 entries with more than 100 features come from 54 names in 18 papers, no name is used by more than two papers, and only two of them (both musk) could be verified; the verified tasks have at most 167 features (median 9).

Table 4. Difficulty Profile of the Verifiable Tasks. A task can carry several flags; entries count how often the corpus uses a task.

	Binary and binarised		One-vs-rest
	Tasks	Entries	Tasks
All	151	594	305
Small disjuncts (rare ≥ 20%)	15 (10%)	45 (8%)	25 (8%)
Outliers (> 20%)	38 (25%)	145 (24%)	40 (13%)
Small sample ($N_{\min}/d < 10$)	103 (68%)	389 (66%)	148 (49%)
None of the three	45 (30%)	197 (33%)	148 (49%)
Imbalance ratio > 9	87	319	148
Noise index, median [P25, P75]	0.44 [0.25, 0.74]		0.30 [0.07, 0.63]

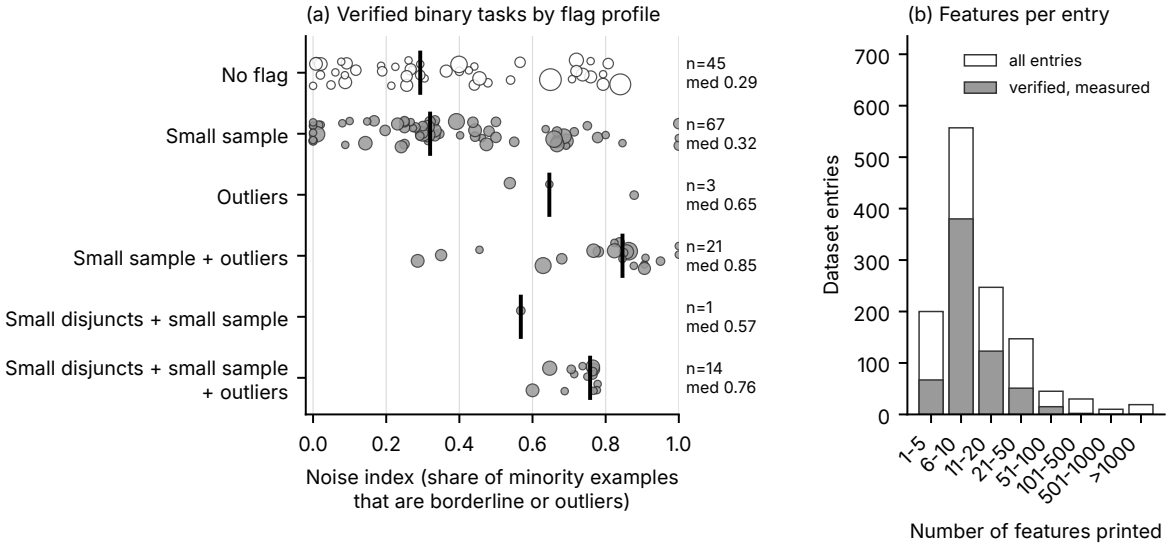


Fig. 1. (a) Noise index of the 151 verifiable binary and binarised tasks, one row per flag profile; marker area grows with the number of entries that use the task, and the vertical bar marks the median. (b) Number of features printed for the 1,255 entries that report it, with the entries that could be verified and measured. The bins are for display only.

The imbalance ratio, the one difficulty-related number most papers print, says little about where the minority examples lie: tasks with a low ratio can be as mixed as tasks with a high one, and a task without any flag is not necessarily easy. Conclusions drawn from the common benchmarks speak mainly for small, low-dimensional tables with few minority examples per feature.

Most papers mix these profiles. Of the 35 papers with at least three verifiable tasks, 32 include both flagged and flag-free tasks, two (COG-OS (Wu et al. 2007) and SMOTE-ENN) use only flag-free tasks and one (MLOS) only flagged ones. Section 1.6 relates the measured profiles to the reported wins.

Table 5. A Reference Collection of Verifiable Two-Class Tasks. One row per selected task; Role: core = most used task in its cell, low/high = lowest/highest noise index in its cell. Profile: S small sample, O outliers, D small disjuncts; Noise: noise index; Entries: corpus entries that use the task.

Task	Role	Profile	IR band	N	$N_{\min}$	d	IR	Noise	Entries
australian	low	none	≤1.5	690	307	14	1.2	0.29	1
bupa	core+high	none	≤1.5	345	145	6	1.4	0.79	6
iris0	low	none	1.5–3	150	50	4	2.0	0.02	2
pima	core	none	1.5–3	768	268	8	1.9	0.65	24
haberman	high	none	1.5–3	306	81	3	2.8	0.84	22
segment0	low	none	3–9	2,308	329	19	6.0	0.02	7
yeast3	core	none	3–9	1,484	163	8	8.1	0.40	13
contraceptive [minority: 2]	high	none	3–9	1,473	333	9	3.4	0.79	1
kr-vs-k-one_vs_fifteen	low	none	>9	2,244	78	6	27.8	0.00	1
shuttle-c0-vs-c4	core	none	>9	1,829	123	9	13.9	0.01	7
yeast-0-2-5-6_vs_3-7-8-9	high	none	>9	1,004	99	8	9.1	0.57	4
heart	core+low	S	≤1.5	270	120	13	1.2	0.33	4
sonar	high	S	≤1.5	208	97	60	1.1	0.40	1
wine [minority: 1]	low	S	1.5–3	178	59	13	2.0	0.00	1
glass0	core	S	1.5–3	214	70	9	2.1	0.44	7
planning-relax	high	S	1.5–3	182	52	12	2.5	0.85	1
musk	low	S	3–9	6,598	1,017	167	5.5	0.01	2
ecoli3	core	S	3–9	336	35	7	8.6	0.69	10
hepatitis (PMLB, N=155)	high	S	3–9	155	32	19	3.8	0.75	2
dermatology-6	low	S	>9	358	20	34	16.9	0.00	2
yeast5	core	S	>9	1,484	44	8	32.7	0.66	13
cleveland-0_vs_4	high	S	>9	177	13	13	12.6	1.00	4
mammography	core+low	O	>9	11,183	260	6	42.0	0.54	5
balance [minority: B]	high	O	>9	625	49	4	11.8	0.88	2
ecoli-0-1-3-7_vs_2-6	low	S+O	>9	281	7	7	39.1	0.29	7
yeast4	core	S+O	>9	1,484	51	8	28.1	0.86	15
ivy-1.4	high	S+O	>9	241	16	20	14.1	1.00	1
breast.cancer	core+low+high	D+S	1.5–3	277	81	9	2.4	0.57	2
heart-disease-processed-cleveland [minority: 4]	low	D+S+O	3–9	303	35	13	7.7	0.71	1
fertility-diagnosis	core+high	D+S+O	3–9	100	12	9	7.3	0.75	1
yeast-1-4-5-8_vs_7	low	D+S+O	>9	693	30	8	22.1	0.60	7
glass2	core	D+S+O	>9	214	17	9	11.6	0.65	9
winequality-red-8_vs_6-7	high	D+S+O	>9	855	18	11	46.5	0.78	1

A reference collection. For readers who need a small benchmark that covers the measured range, Table 5 selects 33 of the 151 tasks by a fixed rule with no threshold of our own. Each combination of flag profile and imbalance-ratio band that occurs (13 combinations) contributes the task that the corpus uses most, for comparability with earlier results, and the tasks with the lowest and the highest noise index, for spread. The collection spans noise indices from 0 to 1, imbalance ratios from 1.1 to 46.5 and sizes from 100 to 11,183 instances, and 15 of its files match the official KEEL table. Its tasks are not independent of one another: it holds five yeast splits and two splits each of ecoli, glass and cleveland, which should be grouped by source when results are tested.

Table 6. Comparisons of the Proposed Method with Plain SMOTE by Metric (29 Papers). A row is one dataset and one metric in one paper; printed values are compared in the direction of the metric, equal printed values are ties, and n.c. counts values not printed as numbers. Scores are not comparable across papers.

Metric	Papers	Rows	Proposed wins	Ties	SMOTE wins	n.c.
AUC	19	403	309	30	64	0
G-mean	19	380	283	29	64	4
F-measure	18	287	217	16	50	4
Recall (sensitivity)	12	193	130	21	42	0
Precision	7	113	62	5	46	0
Accuracy	4	69	62	1	6	0
MCC	3	52	40	0	8	4
AUC-PR	1	7	3	0	0	4
Specificity	1	7	1	0	6	0
All metrics	29	1,511	1,107	102	286	16

Limits of the measurement. The files are copies of the repositories, not downloads from the repositories themselves, whose web sites could not be reached from our machines: the keel-ds and imbalanced-databases packages, two independent GitHub copies of the KEEL standard collection, PMLB, UCI++, the UCI files distributed with Weka, a copy of the tera-PROMISE defect data and single UCI files from a public data-set mirror. For the 72 KEEL files in the official KEEL table, instance and feature counts match in every case, and the two copies of the KEEL standard collection agree in instances and features on all 67 files they share; the other files have not been compared with the repositories. When several copies hold the same data set, each entry added in the second pass is mapped to one copy, KEEL first and PMLB last, because PMLB stores nominal attributes as integer codes, which then enter HVDM as numeric attributes. Rows with missing values are dropped before the neighbours are found; the largest losses are 1,129 of the 3,772 rows of the sick thyroid data and 75 of the 155 rows of hepatitis. The labels are computed in the full feature space, not in any two-dimensional projection. Of the 771 dataset names, 309 have no file in these copies (200 of them name a public source such as UCI or KEEL and could be measured once the files are obtained), 109 are synthetic and 51 have a file whose size differs from every printed one.

1.6 Where Reported Wins over SMOTE Come From

At the paper level, 44 of the 73 papers use plain SMOTE as a baseline, 13 use SMOTE inside a multiclass or ensemble scheme, and 16 do not compare with SMOTE at all. Fewer papers give scores a reader can pair: 29 papers print per-dataset scores for the proposed method and plain SMOTE in the same table. For these papers we compared the printed values on every metric that the table reports for both methods, with the classifier and condition fixed as in Section 1.1; a win means the proposed method's printed value is better in the metric's direction, and equal printed values count as a tie. This gives 1,511 comparisons on 524 datasets (one to six metrics per paper, median three); 16 cannot be decided because a value is not printed as a number. Of the other 1,495 (520 datasets), the proposed method wins 1,107, ties 102 and loses 286, a win in 79.5% of the decided comparisons (95% CI 77.3 to 81.5). Twenty-three of the 29 papers report at least one comparison that SMOTE wins.

Table 6 shows that the margin is not the same on every metric. On AUC, G-mean, F-measure and recall the proposed method wins 76 to 83% of the decided comparisons, and on accuracy and MCC 91 and 83%; on precision the split is only 62 to 46, and on specificity SMOTE wins six of seven. Counting one result per dataset (the majority over its metrics) gives 409 wins, 35 ties and 76 losses on 520 datasets, and the earlier single-metric rule

Table 7. Comparisons with Plain SMOTE by How the Dataset Was Obtained (All Metrics, 29 Papers). Win share is the share of wins among decided comparisons (ties left out), with a 95% Wilson interval. Sixteen comparisons not printed as numbers are left out.

How obtained	Datasets	Papers	Rows	Wins	Ties	Losses	Win share (%)
Original	144	25	425	329	10	86	79.3 (75.1–82.9)
Repository-defined	119	9	315	220	44	51	81.2 (76.1–85.4)
Modified	143	14	495	356	20	119	74.9 (70.9–78.6)
Synthetic	5	1	15	15	0	0	100 (79.6–100)
Cannot be told	105	13	234	184	24	26	87.6 (82.5–91.4)
Not matched to a dataset	4	3	11	3	4	4	42.9 (15.8–75.0)
All	520	29	1,495	1,107	102	286	79.5 (77.3–81.5)

(G-mean, else AUC, else F-measure, else accuracy) gives 521 comparisons, in which the proposed method wins in 402 rows, ties in 39 and loses in 80.

We tested the hypothesis that motivated this analysis: that proposed methods win more often on data the authors modified or generated than on original data. The data do not support it (Table 7). Modified and synthetic data give a win in 75.7% of decided comparisons (371 of 490) against 79.3% on original data. Because the comparisons of one paper are not independent, we estimated the difference with the paper as the unit of clustering: a logistic model with paper-level clustering (generalised estimating equations) gives an odds ratio of 0.68 (95% CI 0.43 to 1.08, $p = 0.10$) for modified or synthetic against original data; a conditional logistic model that compares the two kinds of data only inside the 11 papers that use both and report both outcomes gives 0.61 (0.38 to 0.97, $p = 0.04$); and a mixed model with random effects for paper and for dataset gives 0.75 (0.58 to 0.97). The direction is therefore the opposite of the hypothesis, but it is not stable: with one result per dataset the two clustered estimates are 0.64 ($p = 0.15$) and 0.46 ($p = 0.13$), and under the single-metric rule 0.80 ($p = 0.47$) and 0.61 ($p = 0.31$). Paper by paper, among the 13 papers with both kinds of data, the win share is higher on modified or synthetic data in 3, higher on standard data in 7 and equal in 3 (sign test $p = 0.34$). Repository-defined splits do not differ from original data (odds ratio 1.19, $p = 0.61$), and the comparisons whose data origin cannot be told show the highest raw win share apart from the single paper with synthetic data, a difference that disappears once papers are taken into account (odds ratio 0.99, $p = 0.98$).

The second hypothesis concerned papers that add a single standard dataset to data of another kind. Two papers fit this pattern: MWMOTE (one standard and 19 other datasets) and RUSBoost (Seiffert et al. 2010) (one and 14). Both win on their standard dataset; on their modified datasets and those of unstated origin they win 52 of 65 decided comparisons (80.0%; two MWMOTE datasets that could not be matched to a record are left out), the same share as on such data in the 21 papers that have at least two standard datasets and also report other data (482 of 600, 80.3%). Five papers compare with SMOTE only on standard data and one (INOS) only on modified data. Two papers are too few for a test, and nothing in them suggests that the standard dataset serves a different purpose from the rest.

Measured difficulty says more about where the wins fall than the origin of the data. For the 89 verifiable two-class files (Section 1.5) that appear in these comparisons (22 papers, 440 comparisons), the chance that the proposed method beats SMOTE rises with the noise index: the odds ratio per 0.1 of noise index is 1.13 (95% CI 1.01 to 1.26, $p = 0.03$) with paper-level clustering and 1.17 (1.04 to 1.33, $p = 0.01$) within papers, and 1.19 (1.02 to 1.39, $p = 0.03$) with one result per file and paper. The imbalance ratio shows no such relation (odds ratio 0.91 per doubling, $p = 0.48$), nor do the small-sample, outlier and small-disjunct flags taken one at a time. This analysis is exploratory: it covers the files that could be verified, which are mostly small and low-dimensional.

Table 8. Granularity of the Results Each Paper Reports (73 Papers)

How results are reported	Papers
Per-dataset table	47
Per-dataset, partial (some datasets, metrics or classifiers only)	8
Plots only	5
Averages over all datasets only	5
Ranks only	3
No comparison with other methods	5

Comparisons with SMOTE inside another scheme are kept apart and are not pooled with Tables 6 and 7. Twelve papers contribute 558 comparisons: Static-SMOTE (EMDO (Yao and T.-B. Lin 2021), OVO-SMOTE, SOUP) 79, with 69 wins, 3 ties and 7 losses; SMOTE after one-versus-one decomposition (AMDO (Yang et al. 2018), GDHS (Yan et al. 2025), SHSampler (Zheng et al. 2022)) 148, with 101, 6 and 41; SMOTE applied to all classes (MC-CCR (Koziarski, Woźniak, et al. 2020), MC-RBO (Krawczyk et al. 2020), NROMM) 239, with 231, 0 and 8; and SMOTE combined with another learner (DSRBF (Fernández-Navarro et al. 2011), EasyEnsemble, GAMO) 92, with 71, 2 and 19. Three papers are left out of every comparison set. KNNOR (Islam et al. 2022) prints only means over its 19 datasets (G-score .8363 for KNNOR and .8238 for SMOTE). MC-MBRC (Ma et al. 2024) names SMOTE as a baseline, but the column headers of its per-dataset tables do not say which column is SMOTE. In RBBoost (Díez-Pastor et al. 2015), the scores of its RB-Boost come from the Boosting-family table of its supplement (Supplementary Table 11) and those of its SMOTE ensemble (ESM) from the data-processing-family table (Supplementary Table 3); both report the same protocol (J48 with Laplace smoothing, ensembles of 100, stratified 5×2-fold cross-validation), but the paper compares methods only within each family and never compares RB-Boost with ESM directly. Its 86 datasets, 63 of them repository-defined, are kept in our data for audit.

What the reader should do. Read a reported win over SMOTE together with the metric: precision and specificity often favour SMOTE even when the other metrics do not. Expect wins over SMOTE to be more frequent on data whose minority examples are borderline or isolated, and less informative on clean data. Whether the data were modified does not, on this evidence, inflate the win rate; whether they are described well enough to be identified matters more for the reader (Sections 1.4 and 1.7).

1.7 Can the Reported Results Be Checked?

Of the 57 papers that use SMOTE as a baseline (44 plain, 13 wrapped), 13 give a reader no per-dataset pair of scores in the paper itself: the results are plots (Borderline-SMOTE (H. Han et al. 2005), Safe-Level-SMOTE (Bunkhumpornpat et al. 2009), Cluster-SMOTE (D. Cieslak et al. 2006), k-means SMOTE (Douzas, Bacao, and Last 2018), GLOS (M. Han et al. 2023)), averages (G-SMOTE (Douzas and Bacao 2019), CISO (Yadav and Vijay 2025), SplitBal (Sun et al. 2015), QC-SMOTE (Upman and Gowda 2026)), or ranks (MDO, RBO (Koziarski, Krawczyk, et al. 2019), MC-NCLWO (Tao et al. 2025)), and the review (Hassanat et al. 2022) compares synthetic points rather than classifiers. A fourteenth paper, KNNOR, prints its pair of scores only as a mean over its datasets.

Even where per-dataset tables exist, they do not always cover the datasets a paper lists. In the 43 papers with per-dataset comparisons with SMOTE, 119 of the 952 listed dataset entries (12.5%, 95% CI 10.5 to 14.8) have no row in the table that compares the proposed method with SMOTE, and 12 papers account for all of them, most of all SMOTE-IPF (Sáez et al. 2015) (32 of 41) and Loyalty-SMOTE (30 of 59). The gap is largest for synthetic sets (38 of 47) and for entries whose origin cannot be told (44 of 166), and smallest for modified and repository-defined

data (5 of 180 and 7 of 189). Some of these datasets appear in other results tables of the same paper; what is missing for them is the pair of scores against SMOTE.

Synthetic data are the hardest to check, because a reader can only regenerate them. Of the 19 papers whose evaluation includes author-generated data (120 dataset entries), six give a location for those data (62 entries; for example k-means SMOTE (Douzas, Bacao, and Last 2018) and G-SMOTE (Douzas and Bacao 2019)), two take them from cited work (5 entries), four link only the code of their method (9 entries), and seven say nothing about where the data can be obtained (44 entries, among them the 24 sets of SPIDER2). Ten more papers generate data for an illustration or a side experiment that we did not count as dataset entries; of these, only MC-MBRC (Ma et al. 2024) states that its data are available on request.

Baseline results are also reused across papers. QC-SMOTE copies its baseline numbers from the VS-SMOTE paper (Qiu et al. 2025), and every SMOTE-all value in MC-CCR's Table 6 is identical, to two decimals and on all 20 datasets, to the SMOTE-all column of MC-RBO's Table VI, from the same group and with the same C5.0 setup. The same pattern appears across research groups: the Baseline, SSMOTE, GCS, ABNC, OSMOTE and AMDO columns of EMDO's Table 5 (Yao and T.-B. Lin 2021) repeat, value for value on all nine shared datasets, the MAUC results of AMDO's Table 6 (Yang et al. 2018); EMDO states that it uses the same datasets as AMDO but not that the baseline results were copied. Reusing a baseline is efficient, but unless it is stated, a reader cannot tell whether the comparison was run under identical conditions. Several of these papers place per-dataset results on GitHub or in material outside the paper. Where only ranks are given, the paper's own table can still be informative: in RBO's Table 2 (CART, no added noise) SMOTE has the better mean G-mean rank, and CISO reports itself statistically equivalent to SMOTE.

For these papers the headline claim cannot be falsified from what is published. This does not mean the claims are wrong; it means a reader cannot tell whether an average improvement comes from consistent gains or from large gains on a few datasets.

1.8 Recommendations for Reporting Data

(1) Name the repository and version of every dataset, and whether the full set or a training file was used (Section 1.4).
(2) State which class or classes form the minority and why, in the text and not only in a table (Section 1.3).
(3) Declare every modification (binarisation, class merging, undersampling, noise, dropped attributes) in the data section, and print the size and class counts of the data actually used (Section 1.3).
(4) Give the imbalance-ratio formula, not only its values (Section 1.4).
(5) Report the number of source families alongside the number of datasets, and keep splits of one source together when testing for significance (Section 1.2).
(6) Report per-dataset results for the proposed method and plain SMOTE on every metric used, with standard and modified datasets shown separately (Sections 1.6 and 1.7).
(7) Separate tuning datasets from evaluation datasets, as CISO and LS (D. A. Cieslak and Chawla 2008) do, and run the baselines rather than copying their numbers (Section 1.7).
(8) Report at least one measure of where the minority examples lie (for example the shares of safe, borderline, rare and outlier examples) next to the imbalance ratio (Sections 1.5 and 1.6).

References

L. Abdi and S. Hashemi. Jan. 2016. "To Combat Multi-Class Imbalanced Problems by Means of Over-Sampling Techniques." *IEEE Transactions on Knowledge and Data Engineering*, 28, 1, (Jan. 2016), 238–251. doi:10.1109/TKDE.2015.2458858.

J. Alcalá-Fdez, A. Fernández, J. Luengo, J. Derrac, S. García, L. Sánchez, and F. Herrera. 2011. "KEEL Data-Mining Software Tool: Data Set Repository, Integration of Algorithms and Experimental Analysis Framework." *Journal of Multiple-Valued Logic and Soft Computing*, 17, 2–3, 255–287.

S. Barua, M. M. Islam, X. Yao, and K. Murase. Feb. 2014. "MWMOTE–Majority Weighted Minority Oversampling Technique for Imbalanced Data Set Learning." *IEEE Transactions on Knowledge and Data Engineering*, 26, 2, (Feb. 2014), 405–425. doi:10.1109/TKDE.2012.232.

C. Bunkhumpornpat, K. Sinapiromsaran, and C. Lursinsap. 2009. "Safe-Level-SMOTE: Safe-Level-Synthetic Minority Over-Sampling TEchnique for Handling the Class Imbalanced Problem." In: *Advances in Knowledge Discovery and Data Mining* (Lecture Notes in Computer Science). Springer Berlin Heidelberg, 475–482. doi:10.1007/978-3-642-01307-2_43.

D. Cieslak, N. Chawla, and A. Striegel. 2006. "Combating imbalance in network intrusion datasets." In: *2006 IEEE International Conference on Granular Computing*. Year from the conference title (2006 IEEE GrC); Crossref record has no year field (checked 28 Oct 2026). IEEE, Atlanta, GA, USA, 732–737. doi:10.1109/GRC.2006.1635905.

D. A. Cieslak and N. V. Chawla. Dec. 2008. "Start Globally, Optimize Locally, Predict Globally: Improving Performance on Imbalanced Data." In: *2008 Eighth IEEE International Conference on Data Mining*. (Dec. 2008), 143–152. doi:10.1109/ICDM.2008.87.

J. F. Díez-Pastor, J. J. Rodríguez, C. García-Osorio, and L. I. Kuncheva. Sept. 2015. "Random Balance: Ensembles of variable priors classifiers for imbalanced data." *Knowledge-Based Systems*, 85, (Sept. 2015), 96–111. doi:10.1016/j.knosys.2015.04.022.

G. Douzas and F. Bacao. Oct. 2019. "Geometric SMOTE a geometrically enhanced drop-in replacement for SMOTE." *Information Sciences*, 501, (Oct. 2019), 118–135. doi:10.1016/j.ins.2019.06.007.

G. Douzas, F. Bacao, and F. Last. Oct. 2018. "Improving imbalanced learning through a heuristic oversampling method based on k-means and SMOTE." *Information Sciences*, 465, (Oct. 2018), 1–20. doi:10.1016/j.ins.2018.06.056.

A. Fernández, S. García, M. J. del Jesus, and F. Herrera. 2008. "A study of the behaviour of linguistic fuzzy rule based classification systems in the framework of imbalanced data-sets." *Fuzzy Sets and Systems*, 159, 18, 2378–2398. Used: sec. 5, IR bands 1.5–3, 3–9, >9 (p. 14 of the in-press version). doi:10.1016/j.fss.2007.12.023.

A. Fernández, V. López, M. Galar, M. J. del Jesus, and F. Herrera. Apr. 2013. "Analysing the classification of imbalanced data-sets with multiple classes: Binarization techniques and ad-hoc approaches." *Knowledge-Based Systems*, 42, (Apr. 2013), 97–110. doi:10.1016/j.knosys.2013.01.018.

F. Fernández-Navarro, C. Hervás-Martínez, and P. A. Gutiérrez. Aug. 2011. "A dynamic over-sampling procedure based on sensitivity for multi-class problems." *Pattern Recognition*, 44, 8, (Aug. 2011), 1821–1833. doi:10.1016/j.patcog.2011.02.019.

L. P. F. Garcia. N.d. *UCI++: A Huge Collection of Preprocessed Datasets for Supervised Classification Problems in ARFF Format*. GitHub repository lpfgarcia/ucipp and Zenodo. (). doi:10.5281/zenodo.13748.

H. Han, W.-Y. Wang, and B.-H. Mao. 2005. "Borderline-SMOTE: A New Over-Sampling Method in Imbalanced Data Sets Learning." In: *Advances in Intelligent Computing* (Lecture Notes in Computer Science). Springer Berlin Heidelberg, 878–887. doi:10.1007/11538059_91.

M. Han, H. Guo, J. Li, and W. Wang. June 2023. "Global-local information based oversampling for multi-class imbalanced data." *International Journal of Machine Learning and Cybernetics*, 14, 6, (June 2023), 2071–2086. doi:10.1007/s13042-022-01746-w.

A. B. Hassanat, A. S. Tarawneh, G. A. Altarawneh, and A. Almuhaimeed. 2022. *Stop Oversampling for Class Imbalance Learning: A Critical Review*. Version v2 (8 Jun 2022), the version read for this survey (title page of the corpus PDF). Also a Research Square preprint, doi 10.21203/rs.3.rs-1336037/v1, and a journal version with a different title: Tarawneh et al., IEEE Access 10:47643–47660, 2022, doi 10.1109/ACCESS.2022.3169512. arXiv: 2202.03579 [cs.LG].

A. Islam, S. B. Belhaouari, A. U. Rehman, and H. Bensmail. Jan. 2022. "KNNOR: An oversampling technique for imbalanced datasets." *Applied Soft Computing*, 115, (Jan. 2022), 108288. doi:10.1016/j.asoc.2021.108288.

A. K. Jain, R. P. W. Duin, and J. Mao. 2000. "Statistical pattern recognition: a review." *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 22, 1, 4–37. Used: sec. 3, n/d > 10 samples per class per feature, attributed to their ref. [80]. doi:10.1109/34.824819.

T. Jo and N. Japkowicz. 2004. "Class Imbalances versus Small Disjuncts." *ACM SIGKDD Explorations Newsletter*, 6, 1, 40–49. doi:10.1145/1007730.1007737.

M. Jureczko and L. Madeyski. 2010. "Towards Identifying Software Project Clusters with Regard to Defect Prediction." In: *Proceedings of the 6th International Conference on Predictive Models in Software Engineering (PROMISE '10)*. ACM, 1–10. doi:10.1145/1868328.1868342.

M. Kelly, R. Longjohn, and K. Nottingham. 2026. *The UCI Machine Learning Repository*. Retrieved Nov. 8, 2026 from https://archive.ics.uci.edu.

M. Koziarski, B. Krawczyk, and M. Woźniak. May 2019. "Radial-Based oversampling for noisy imbalanced data classification." *Neurocomputing*, 343, (May 2019), 19–33. doi:10.1016/j.neucom.2018.04.089.

M. Koziarski, M. Woźniak, and B. Krawczyk. Sept. 2020. "Combined Cleaning and Resampling algorithm for multi-class imbalanced data with label noise." *Knowledge-Based Systems*, 204, (Sept. 2020), 106223. doi:10.1016/j.knosys.2020.106223.

B. Krawczyk, M. Koziarski, and M. Wozniak. Aug. 2020. "Radial-Based Oversampling for Multiclass Imbalanced Data Classification." *IEEE Transactions on Neural Networks and Learning Systems*, 31, 8, (Aug. 2020), 2818–2831. doi:10.1109/TNNLS.2019.2913673.

G. Lemaître, F. Nogueira, and C. K. Aridas. 2017. "Imbalanced-learn: A Python Toolbox to Tackle the Curse of Imbalanced Datasets in Machine Learning." *Journal of Machine Learning Research*, 18, 17, 1–5. https://jmlr.org/papers/v18/16-365.html.

V. López, A. Fernández, S. García, V. Palade, and F. Herrera. 2013. "An Insight into Classification with Imbalanced Data: Empirical Results and Current Trends on Using Data Intrinsic Characteristics." *Information Sciences*, 250, 113–141. doi:10.1016/j.ins.2013.07.007.

T. Ma, S. Lu, and C. Jiang. Apr. 2024. "A membership-based resampling and cleaning algorithm for multi-class imbalanced overlapping data." *Expert Systems with Applications*, 240, (Apr. 2024), 122565. doi:10.1016/j.eswa.2023.122565.

K. Napierala and J. Stefanowski. 2016. “Types of minority class examples and their influence on learning classifiers from imbalanced data.” *Journal of Intelligent Information Systems*, 46, 3, 563–597. Online 9 Jul 2015. Used: sec. 5.2.1 labelling rule (k=5, incl. condition for rare), sec. 6.2 “about 20% or more” rare, “more than 20%” outlier. doi:10.1007/s10844-015-0368-1.

J. Ortigosa-Hernández, I. Inza, and J. A. Lozano. Oct. 2017. “Measuring the class-imbalance extent of multi-class problems.” *Pattern Recognition Letters*, 98, (Oct. 2017), 32–38. doi:10.1016/j.patrec.2017.08.002.

A. S. Palli, J. Jaafar, M. A. Hashmani, H. M. Gomes, and A. R. Gilal. 2022. “A Hybrid Sampling Approach for Imbalanced Binary and Multi-Class Data Using Clustering Analysis.” *IEEE Access*, 10, 118639–118653. doi:10.1109/ACCESS.2022.3218463.

F. Qiu, F. Yang, Y. Xiong, T. Chen, P. Xiao, W. Zheng, X. Chen, and K. Zhou. Nov. 2025. “VS-SMOTE: Leveraging high-value spaces to balance noise control and data diversity for class imbalanced learning.” *Neurocomputing*, 654, (Nov. 2025), 131321. doi:10.1016/j.neucom.2025.131321.

J. D. Romano et al.. 2022. “PMLB v1.0: An Open-Source Dataset Collection for Benchmarking Machine Learning Methods.” *Bioinformatics*, 38, 3, 878–880. doi:10.1093/bioinformatics/btab727.

J. A. Sáez, J. Luengo, J. Stefanowski, and F. Herrera. 2015. “SMOTE–IPF: Addressing the Noisy and Borderline Examples Problem in Imbalanced Classification by a Re-Sampling Method with Filtering.” *Information Sciences*, 291, 184–203. doi:10.1016/j.ins.2014.08.051.

C. Seiffert, T. M. Khoshgoftaar, J. Van Hulse, and A. Napolitano. Jan. 2010. “RUSBoost: A Hybrid Approach to Alleviating Class Imbalance.” *IEEE Transactions on Systems, Man, and Cybernetics - Part A: Systems and Humans*, 40, 1, (Jan. 2010), 185–197. doi:10.1109/TSMCA.2009.2029559.

Z. Sun, Q. Song, X. Zhu, H. Sun, B. Xu, and Y. Zhou. May 2015. “A novel ensemble method for classifying imbalanced data.” *Pattern Recognition*, 48, 5, (May 2015), 1623–1637. doi:10.1016/j.patcog.2014.11.014.

L. Tao, Y. Liang, S. Wu, J. Qian, and F. Zhan. 2025. “MC-NCLWO: Newton’s Cooling Law-Based Oversampling Algorithm for Multi-Class Imbalanced Datasets With Feature Noise.” *IEEE Access*, 13, 152620–152636. doi:10.1109/ACCESS.2025.3604051.

P. Upman and S. N. Gowda. June 23, 2026. *QC-SMOTE: Quality-Controlled SMOTE for Imbalanced Classification*. (June 23, 2026). arXiv: 2606.24625 (cs.LG).

J. Wang and N. Awang. 2024. “MKC-SMOTE: A Novel Synthetic Oversampling Method for Multi-Class Imbalanced Data Classification.” *IEEE Access*, 12, 196929–196938. doi:10.1109/ACCESS.2024.3521120.

D. R. Wilson and T. R. Martinez. 1997. “Improved Heterogeneous Distance Functions.” *Journal of Artificial Intelligence Research*, 6, 1–34. Used: HVDM, Eq. 11–13; N2 normalized_vdm for nominal attributes (sec. 3.2). doi:10.1613/jair.346.

J. Wu, H. Xiong, P. Wu, and J. Chen. Aug. 12, 2007. “Local decomposition for rare class analysis.” In: *Proceedings of the 13th ACM SIGKDD international conference on Knowledge discovery and data mining*. (Aug. 12, 2007), 814–823. doi:10.1145/1281192.1281279.

P. Yadav and V. Vijay. Jan. 27, 2025. *Certified Interpolation Oversampling: Per-Instance Safety Guarantees for Imbalanced Learning*. (Jan. 27, 2025). arXiv: 2501.15790 (cs.LG).

Y. Yan, Y. Lv, S. Han, C. Yu, and P. Zhou. July 2025. “GDHS: An efficient hybrid sampling method for multi-class imbalanced data classification.” *Neurocomputing*, 637, (July 2025), 130088. doi:10.1016/j.neucom.2025.130088.

X. Yang, Q. Kuang, W. Zhang, and G. Zhang. Sept. 2018. “AMDO: An Over-Sampling Technique for Multi-Class Imbalanced Problems.” *IEEE Transactions on Knowledge and Data Engineering*, 30, 9, (Sept. 2018), 1672–1685. doi:10.1109/TKDE.2017.2761347.

L. Yao and T.-B. Lin. Oct. 4, 2021. “Evolutionary Mahalanobis Distance-Based Oversampling for Multi-Class Imbalanced Data Classification.” *Sensors*, 21, 19, (Oct. 4, 2021), 6616. doi:10.3390/s21196616.

Z. Zheng, Y. Yan, Y. Zhang, and Y. Zhang. Oct. 13, 2022. “Combating Mutuality with Difficulty Factors in Multi-class Imbalanced Data: A Similarity-based Hybrid Sampling.” In: *2022 IEEE 9th International Conference on Data Science and Advanced Analytics (DSAA)*. (Oct. 13, 2022), 1–10. doi:10.1109/DSAA54385.2022.10032369.

T. Zhu, Y. Lin, and Y. Liu. Dec. 2017. “Synthetic minority oversampling technique for multiclass imbalance problems.” *Pattern Recognition*, 72, (Dec. 2017), 327–340. doi:10.1016/j.patcog.2017.07.024.

T. Zhu, X. Liu, and E. Zhu. June 2023. “Oversampling with Reliably Expanding Minority Class Regions for Imbalanced Data Learning.” *IEEE Transactions on Knowledge and Data Engineering*, 35, 6, (June 2023), 6167–6181. Volume, issue and pages from the printed header and page numbers of the published PDF in paper/Multiclass (checked 29 Oct 2026). doi:10.1109/TKDE.2022.3171706.